



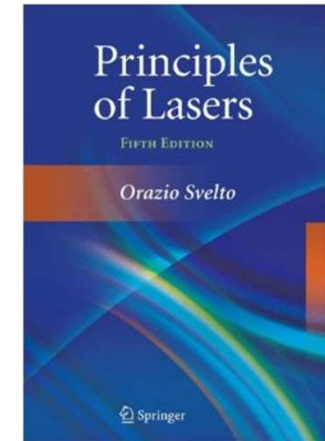
南方科技大学  
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

# 《激光原理与技术》

## Principles and Technologies of Lasers

### Ch3-2

### Resonant Interaction between Electromagnetic Field & Matter



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# Chapter 3: Resonant Interaction between Electromagnetic Field & Matter

## ❖ 3.2 Line broadening mechanisms

- Modification of SP, STE and STA considering the line broadening

## ❖ 3.3 Rate equation of typical lasers

- Rate equation for three-level system (single-mode)
- Rate equation for four-level system (single-mode)
- Rate equation for multi-mode resonance

## 3.2 Modification of the rate equation of SP, STA, and STE

Spectral Line function  $\tilde{g}(\nu, \nu_0)$   $\Rightarrow$  Transition possibility at different frequencies

$$P(\nu) = n_2 h \nu_0 \underline{A_{21} \tilde{g}(\nu, \nu_0)} = n_2 h \nu_0 \underline{A_{21}(\nu)} \quad A_{21}(\nu) = A_{21} \tilde{g}(\nu, \nu_0)$$

$$B_{21} = \frac{c^3}{8\pi h \nu^3} A_{21} \quad A_{21}(\nu) = A_{21} \tilde{g}(\nu, \nu_0) \quad B_{21} = \frac{c^3}{8\pi h \nu^3} \frac{A_{21}(\nu)}{\tilde{g}(\nu, \nu_0)}$$

$$\left( \frac{dn_{21}}{dt} \right)_{sp} = A_{21} n_2 \quad \text{or} \quad B_{21} \tilde{g}(\nu, \nu_0) = B_{21}(\nu) = \frac{c^3}{8\pi h \nu^3} A_{21}(\nu)$$

$$W_{12} = B_{12} \rho_\nu \quad W_{21} = B_{21} \rho_\nu$$

$$\left( \frac{dn_{21}}{dt} \right)_{st} = W_{21} n_2$$

$$\left( \frac{dn_{21}}{dt} \right)_{sp} = \int_{-\infty}^{+\infty} n_2 A_{21}(\nu) d\nu = n_2 A_{21}$$

**Line broadening does NOT influent SP**

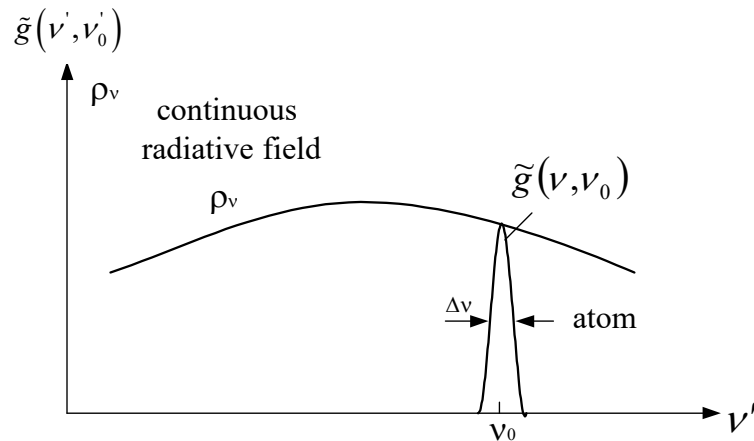
$$W_{21}(\nu) = B_{21}(\nu) \rho_\nu = B_{21} \tilde{g}(\nu, \nu_0) \rho_\nu$$



Modified expression  $\left( \frac{dn_{21}}{dt} \right)_{st} = \int_{-\infty}^{+\infty} n_2 W_{21}(\nu) d\nu = n_2 B_{21} \int_{-\infty}^{+\infty} \tilde{g}(\nu, \nu_0) \rho_\nu d\nu$

Two cases for  $\left(\frac{dn_{21}}{dt}\right)_{st} = \int_{-\infty}^{+\infty} n_2 W_{21}(\nu) d\nu = n_2 B_{21} \int_{-\infty}^{+\infty} \tilde{g}(\nu, \nu_0) \rho_\nu d\nu$

(1) The interaction of atoms with continuous radiation

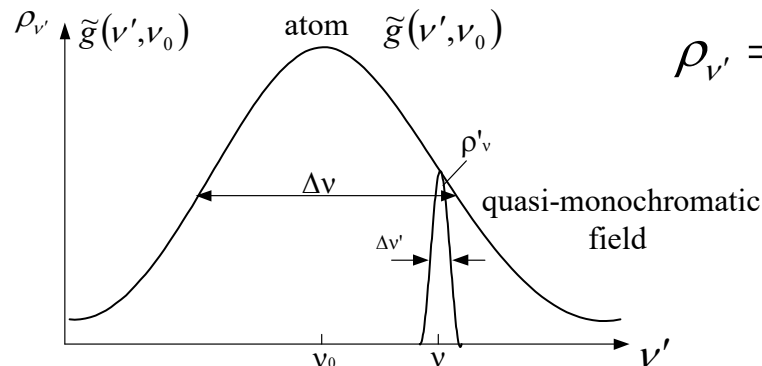


$$\Delta \nu' \gg \Delta \nu$$

$$\left(\frac{dn_{21}}{dt}\right)_{st} = n_2 B_{21} \int_{-\infty}^{+\infty} \tilde{g}(\nu, \nu_0) \rho_{\nu_0} d\nu = n_2 B_{21} \rho_{\nu_0}$$

$$W_{21}(\nu) = B_{21} \rho_{\nu_0} \quad W_{12}(\nu) = B_{12} \rho_{\nu_0}$$

(2) The interaction of atoms with quasi-monochromatic radiation  $\Delta \nu' \ll \Delta \nu$



$$\rho_{\nu'} = \rho \delta(\nu' - \nu) \quad \int_{-\infty}^{+\infty} \rho_{\nu'} d\nu' = \int_{-\infty}^{+\infty} \rho \delta(\nu' - \nu) d\nu' = \rho$$

$$\left(\frac{dn_{21}}{dt}\right)_{st} = n_2 B_{21} \int_{-\infty}^{+\infty} \tilde{g}(\nu, \nu_0) \rho \delta(\nu' - \nu) d\nu' = n_2 B_{21} \tilde{g}(\nu, \nu_0) \rho$$

$$W_{21} = B_{21} \tilde{g}(\nu, \nu_0) \rho \quad W_{12} = B_{12} \tilde{g}(\nu, \nu_0) \rho$$

**Physical Meaning:** due to the line broadening, the light with frequency  $\nu$  does not have to be exactly the same as the center frequency  $\nu_0$  to have stimulated recombination. Instead, the frequency in a certain range near  $\nu=\nu_0$  can generate stimulated emission.

Other expression of STE and STA possibility

photon density of mode  $l$

mode density

$$n_\nu = 8\pi\nu^2/c^3$$

$$\because \rho = N_l h\nu \quad A_{21}/B_{21} = 8\pi h\nu^3/c^3 = n_\nu h\nu = n_\nu \frac{\rho}{N_l} \Rightarrow B_{21} = \frac{N_l A_{21}}{n_\nu \rho}$$

$$W_{21} = B_{21} \tilde{g}(\nu, \nu_0) \rho = \frac{A_{21}}{n_\nu} \tilde{g}(\nu, \nu_0) N_l = \sigma_{21}(\nu, \nu_0) \nu N_l$$

$$W_{12} = B_{12} \tilde{g}(\nu, \nu_0) \rho = \frac{f_2}{f_1} \frac{A_{12}}{n_\nu} \tilde{g}(\nu, \nu_0) N_l = \sigma_{12}(\nu, \nu_0) \nu N_l$$

Emission cross-section  $\sigma_{21}(\nu, \nu_0) = \frac{A_{21}\nu^2}{8\pi\nu_0^2} \tilde{g}(\nu, \nu_0)$

Absorption cross-section  $\sigma_{12}(\nu, \nu_0) = \frac{f_2}{f_1} \frac{A_{12}\nu^2}{8\pi\nu_0^2} \tilde{g}(\nu, \nu_0)$

**Emission cross-section reaches its maximum at central frequency**

Emission cross-section of  
homogenous broadening at  
central frequency

$$\sigma_{21}(\nu_0, \nu_0) = \frac{A_{21}\nu^2}{4\pi^2\nu_0^2\Delta\nu_H}$$

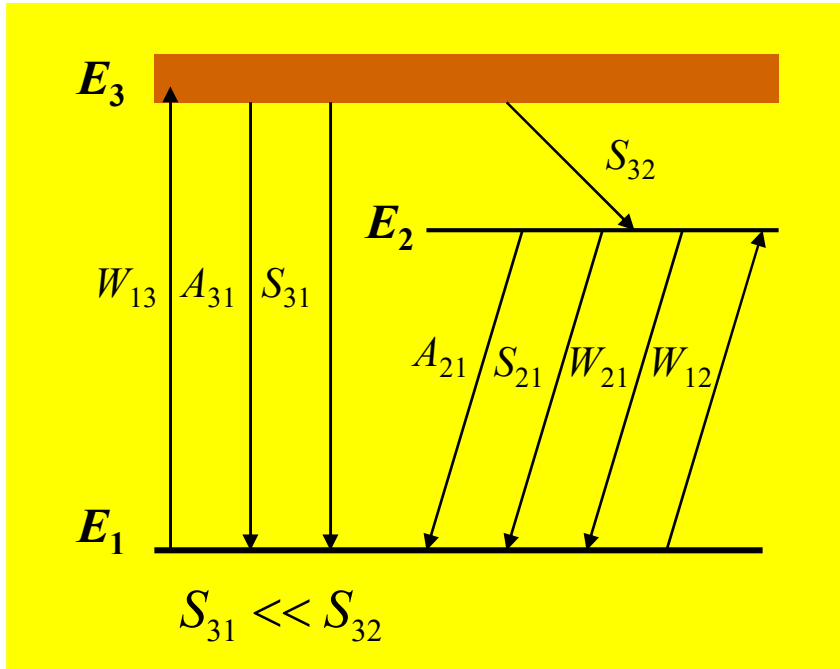
Emission cross-section of  
inhomogeneous broadening at  
central frequency

$$\sigma_{21}(\nu_0, \nu_0) = \frac{\sqrt{\ln 2} A_{21}\nu^2}{4\pi^{3/2}\nu_0^2\Delta\nu_D}$$

# 3.3 Rate equation of typical lasers

四能级泵浦阈值更低

❖ Rate equation for three-level system (ruby laser)



$$\begin{cases} \frac{dn_3}{dt} = n_1 W_{13} - n_3 (S_{32} + A_{31} + \cancel{S_{31}}) \\ \frac{dn_2}{dt} = n_1 W_{12} - n_2 W_{21} - n_2 (A_{21} + S_{21}) + n_3 S_{32} \\ n_1 + n_2 + n_3 = n \end{cases}$$

Consider the cavity mode  $l$

$$\frac{dN_l}{dt} = n_2 W_{21} - n_1 W_{12} - \frac{N_l}{\tau_{Rl}}$$

The average photon lifetime

$$I_m = I_0 e^{-2\delta m}$$

$m = \frac{t}{(2L'/c)}$

optical length

$$I_m = I_0 e^{-\frac{2\delta c t}{2L'}} \Rightarrow \frac{1}{\tau_R} \Rightarrow \boxed{\tau_R = L'/\delta c}$$

$$I(t) = I_0 e^{-\frac{t}{\tau_R}}$$

$$W_{21} = B_{21} \tilde{g}(\nu, \nu_0) \rho = \frac{A_{21}}{n_\nu} \tilde{g}(\nu, \nu_0) N_l = \sigma_{21}(\nu, \nu_0) \nu N_l$$

$$W_{12} = B_{12} \tilde{g}(\nu, \nu_0) \rho = \frac{f_2}{f_1} \frac{A_{12}}{n_\nu} \tilde{g}(\nu, \nu_0) N_l = \sigma_{12}(\nu, \nu_0) \nu N_l$$

❖ Rate equation for three-level system

$$\left\{ \begin{array}{l} \frac{dn_3}{dt} = n_1 W_{13} - n_3 (S_{32} + A_{31}) \\ \frac{dn_2}{dt} = - \left( n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l - n_2 (A_{21} + S_{21}) + n_3 S_{32} \\ n_1 + n_2 + n_3 = n \\ \frac{dN_l}{dt} = \left( n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l - \frac{N_l}{\tau_{Rl}} \end{array} \right.$$

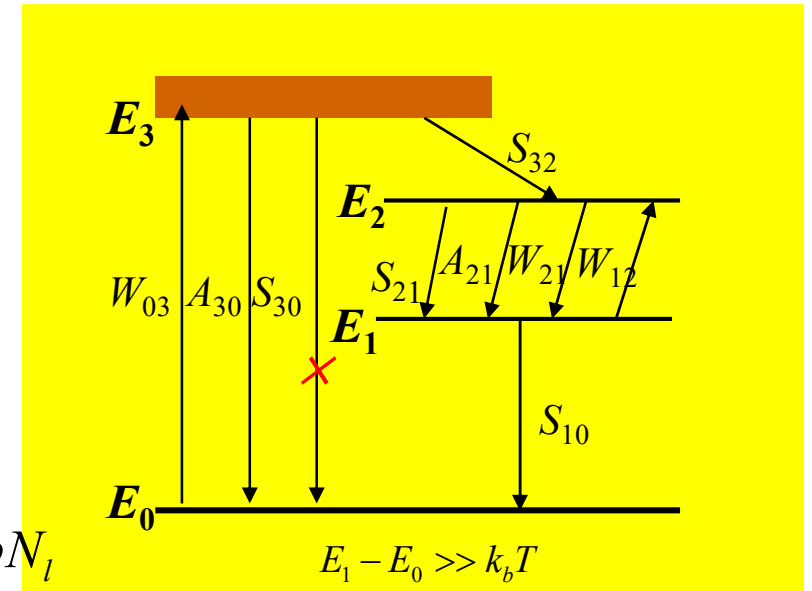


❖ Rate equation for four-level system (Nd:YAG)

$$\frac{dn_3}{dt} = n_0 W_{03} - n_3 (S_{32} + A_{30})$$

$$\frac{dn_2}{dt} = n_3 S_{32} - n_2 (S_{21} + A_{21}) + n_1 W_{12} - n_2 W_{21}$$

$$= n_3 S_{32} - n_2 (S_{21} + A_{21}) - \left( n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l$$



$$\frac{dn_0}{dt} = n_1 S_{10} - n_0 W_{03} + n_3 A_{30}$$

$$W_{21} = \sigma_{21}(\nu, \nu_0) \nu N_l$$

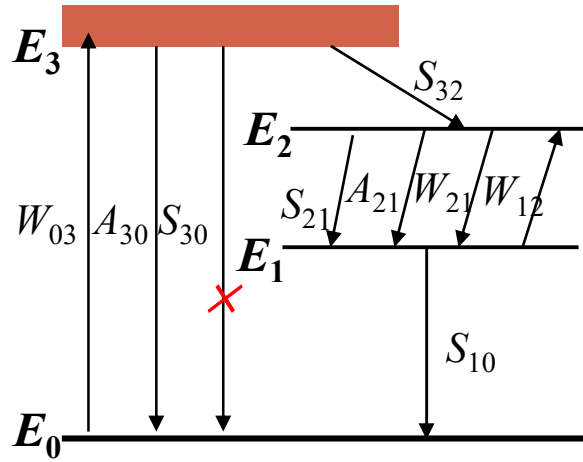
$$\begin{aligned} \frac{dN_l}{dt} &= n_2 W_{21} - n_1 W_{12} - \frac{N_l}{\tau_{Rl}} \\ &= \left( n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l - \frac{N_l}{\tau_{Rl}} \end{aligned}$$

$$n_0 + n_1 + n_2 + n_3 = n$$

❖ Rate equation for four-level system

$$\left\{ \begin{array}{l} \frac{dn_3}{dt} = n_0 W_{03} - n_3 (S_{32} + A_{30}) \\ \frac{dn_2}{dt} = n_3 S_{32} - n_2 (S_{21} + A_{21}) - \left( n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l \\ \frac{dn_0}{dt} = n_1 S_{10} - n_0 W_{03} + n_3 A_{30} \\ n_0 + n_1 + n_2 + n_3 = n \\ \frac{dN_l}{dt} = \left( n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l - \frac{N_l}{\tau_{Rl}} \end{array} \right.$$

- Rate equation of laser is established according to the specific physical processes
- The rate equation of a laser can have different expressions

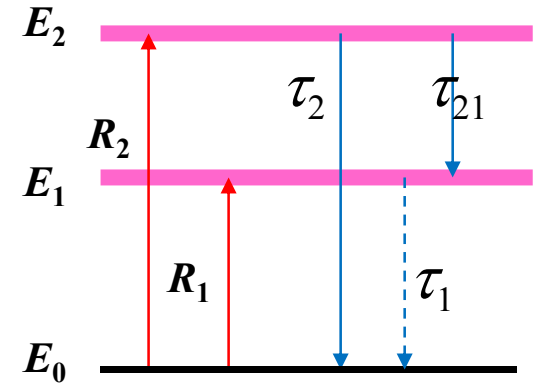


$$\frac{dn_2}{dt} = n_3 S_{32} - n_2 (S_{21} + A_{21}) + n_1 W_{12} - n_2 W_{21}$$

$$\frac{dn_1}{dt} = n_2 (S_{21} + A_{21}) - n_1 W_{12} + n_2 W_{21} - n_1 S_{10}$$

$$\frac{dn_2}{dt} = R_2 - \frac{n_2}{\tau_2} - \left( n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l$$

$$\frac{dn_1}{dt} = R_1 - \frac{n_1}{\tau_1} + \frac{n_2}{\tau_{21}} + \left( n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l$$



$R_1$  and  $R_2$  is the unit time to pump carriers from ground state to  $E_1$  and  $E_2$ .

$\tau_1$  and  $\tau_2$  is the lifetime of energy level  $E_1$  and  $E_2$

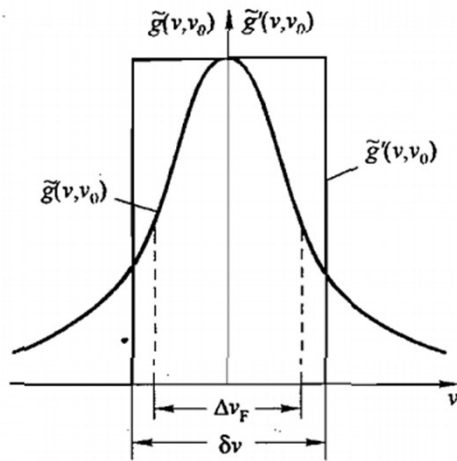
$\tau_{21}$  is the lifetime of spontaneous emission from  $E_2 \rightarrow E_1$

## ❖ Rate equation for multi-mode resonance

$$l = l_1, l_2 \cdots l_n \quad \nu = \nu_{l_1}, \nu_{l_2} \cdots \nu_{l_n} \quad N_l = N_{l_1}, N_{l_2} \cdots N_{l_n}$$

$$\frac{dn_2}{dt} = n_3 S_{32} - n_2 (A_{21} + S_{21}) - \sum_l \left( n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu_l, \nu_0) \nu N_l$$

$$\frac{dN_l}{dt} = \left( n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21}(\nu, \nu_0) \nu N_l - \frac{N_l}{\tau_{Rl}} \rightarrow \frac{dN_{l_1}}{dt} = \cdots, \cdots \frac{dN_{l_n}}{dt} = \cdots$$



### To simplify

- No need to consider the difference between different modes
- The difference between diffraction loss can be ignored



Line function treated as rectangular shape



The mode loss and lifetime of modes are the same

$$\tilde{g}'(\nu, \nu_0) = \tilde{g}(\nu_0, \nu_0)$$

$$\delta\nu = 1/\tilde{g}(\nu_0, \nu_0)$$

$$\sum_l \sigma_{21}(\nu_l, \nu_0) \nu N_l \rightarrow \sigma_{21} \nu \sum_l N_l \rightarrow \sigma_{21} \nu N$$

## Simplified rate equation for multi-mode resonance

$$\frac{dn_3}{dt} = n_0 W_{03} - \frac{n_3 S_{32}}{\eta_1} \quad \eta_1 = \frac{S_{32}}{S_{32} + A_{30}} \quad E_3 \rightarrow E_2 \text{ quantum efficiency (pumping efficiency)}$$

*(Note: In the original image,  $n_3 S_{32}$  and  $n_3(S_{32} + A_{30})$  are circled in red, with a red arrow pointing from the first to the second.)*

$$\frac{dn_2}{dt} = - \left( n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21} \nu N - \frac{n_2 A_{21}}{\eta_2} + n_3 S_{32} \quad \eta_2 = \frac{A_{21}}{A_{21} + S_{21}} \quad E_2 \rightarrow E_1 \text{ fluoresce quantum efficiency (quantum yield)}$$

*(Note: In the original image,  $n_2 A_{21}$  and  $n_2(A_{21} + S_{21})$  are circled in red, with a red arrow pointing from the first to the second. The term  $\sigma_{21} \nu N$  is underlined in red.)*

$$\frac{dn_0}{dt} = n_1 S_{10} - n_0 W_{03} + n_3 A_{30}$$

*(Note: In the original image,  $n_3 A_{30}$  is crossed out with a red diagonal line.)*

$$\frac{dN}{dt} = \left( n_2 - \frac{f_2}{f_1} n_1 \right) \sigma_{21} \nu N - \frac{N}{\tau_R}$$

$N$ : total number of photons from all modes

$$n_0 + n_1 + n_2 + n_3 = n$$

Total emitted photons

Quantum Efficiency  $\eta_F = \eta_1 \eta_2 = \frac{\text{Photon numbers absorbed by the materials through optical pumping}}{\text{Total emitted photons}}$